TERRASIGHT AI

PID_2024_CSE_2_39

Mini Project / 5th Sem/ BCS554

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SUPERVISOR NAME:

[Ms. Manisha]

ABSTRACT

- Urban waste management is reactive — manual complaints and fixed sweeping routes let illegal dumping go undetected.
- Terrasight AI is a computer-vision and spatial-intelligence platform for municipal sanitation departments.
- A fine-tuned YOLOv8 model detects waste and computes a Surface-Area Severity Score (0–100%).
- DBSCAN clusters geo-tagged citizen reports (via EXIF GPS) into priority cleanup hotspots.
- Sanitation officers get a color-coded, interactive GIS dashboard ranking dump-sites by severity.
- Replaces manual guesswork with data-driven dispatch of sanitation vehicles.

OBJECTIVE

- Detect and classify waste types from field/citizen photos using a fine-tuned YOLOv8 model.
- Compute an objective, repeatable Severity Score (0–100%) from bounding-box coverage.
- Extract GPS coordinates from photo EXIF metadata automatically, with manual fallback.
- Cluster nearby reports into unified dump-site hotspots using DBSCAN + Haversine distance.
- Visualize hotspots on a color-coded, interactive GIS dashboard for sanitation officers.
- Enable data-driven dispatch of sanitation vehicles based on real-time severity ranking.

SCOPE of the Problem Statement

- Problem: urban waste management is largely reactive, relying on manual complaints and fixed truck routes.
- Illegal dumping blackspots accumulate silently in plots, roadside edges, and near drains.
- No severity prioritization — a small litter pile is treated the same as a massive dump.
- Target Users: municipal corporations, Nagar Nigams, and urban sanitation departments.
- Technical Scope: YOLOv8 computer vision, DBSCAN clustering, FastAPI backend, React-Leaflet GIS frontend.
- Real-World Usage: severity-ranked cleanup routing in place of fixed daily sweeps.

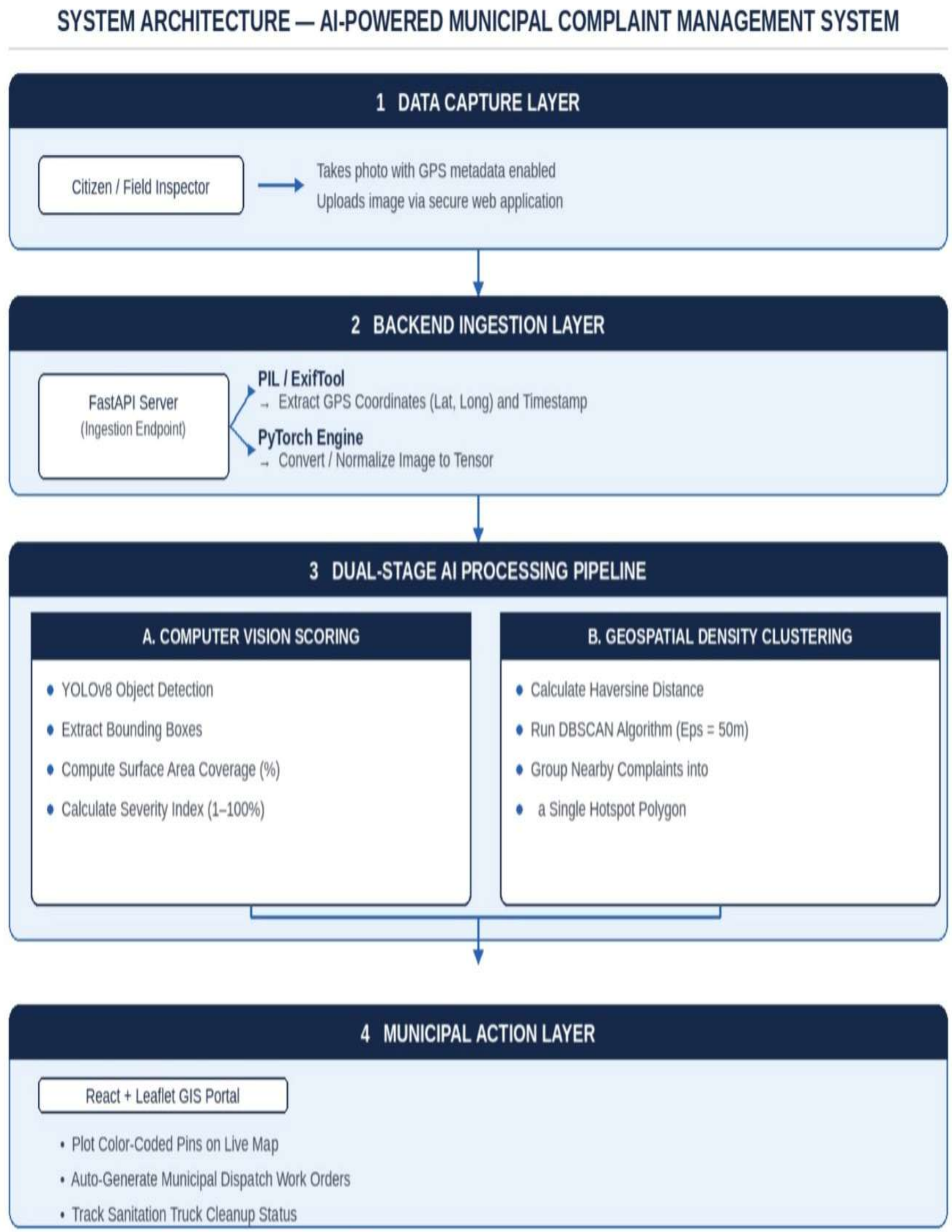
RELATED WORK

- **Traditional Civic Portals:** Standard municipal helplines rely on manual text and pin-drops, resulting in imprecise location data, zero severity metrics, and high administrative overhead.
- **Basic Image Classification:** Existing CNN research only performs binary detection (Garbage vs. No Garbage) without quantifying total surface area coverage or waste hazard levels.
- **Manual Location Input:** Conventional web forms require users to type addresses manually instead of reading raw image metadata, increasing user friction and location errors.
- **Duplicate Work Orders:** Current portals process every upload as an isolated complaint, creating multiple redundant cleanup tickets for the same dump-site.
- **The TerraSight AI Advancement:** Integrates **YOLOv8 bounding-box area math** for automated 0–100% severity scoring with **EXIF GPS extraction and DBSCAN clustering** to aggregate nearby reports into single GIS hotspots.

Applications

- Municipal corporations & Nagar Nigams — automated illegal dump-site monitoring.
- Urban sanitation departments — severity-based dispatch of cleanup vehicles.
- Field inspectors & citizens — quick photo-based reporting of waste sites.
- Sanitation officer dashboards — real-time GIS hotspot visualization for decision-making.

System architecture



Limitations

- Severity scoring depends on photo quality, angle, and framing at capture time.
- Requires GPS EXIF metadata; a manual pin-drop fallback is needed when it is missing.
- Category Hazard Weights (wc) are fixed values (Organic = 1.2, Plastic = 1.0, Dry Leaves = 0.7).
- DBSCAN clustering needs a distance threshold, tuned for dense vs. sparse city zones.

Conclusion and Future Scope

CONCLUSION

- delivers a two-stage AI pipeline combining YOLOv8 detection with DBSCAN spatial clustering.
- converts citizen photos into a quantifiable, repeatable Severity Score.
- aggregates redundant reports into unified, dispatch-ready hotspots on a live GIS map.

FUTURE SCOPE

- mobile app with push notifications for citizen reporters and field officers.
- automated vehicle-routing optimization layered on hotspot data.
- integration with municipal ERP / grievance systems for closed-loop resolution.

Annexure 10 B (Duly Signed by Supervisor)

ANNEXURE X (B) (PROJECT REVIEW REPORT)

ANNEXURE X (B)

ABES Engineering College, Ghaziabad

Department Computer Science and Engineering

Project Review Report (Session: 2026 - 2027 **)**

PID: 2024 CSE 2 39

- **Project Title:** TerraSight AI
- **Project Supervisor:** Ms. Manisha
- **Particular of Student(s):**

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- **Remarks from Supervisor** Project topic has been finalized.
- **Remarks from DPC**

The Progress of the Group is found to be (Satisfactory/ Unsatisfactory).

Signature of DPC Member

Date:

Manisha

Supervisor's Signature

Date:

References

- Ultralytics YOLOv8 — object detection framework used for waste classification. [Full citation — Information Required]
- DBSCAN (Density-Based Spatial Clustering of Applications with Noise) — spatial clustering algorithm. [Full citation — Information Required]
- TACO (Trash Annotations in Context) dataset — open-source annotated waste image dataset. [Full citation — Information Required]
- TrashNet dataset — used for waste category classification verification. [Full citation — Information Required]
- OpenStreetMap API — basemap provider for the GIS dashboard. [Full citation — Information Required]
- FastAPI, React.js, React-Leaflet — framework/library documentation. [Full citation — Information Required]